

CADEMAS-ML

CADEMAS Instantiation

Dataset(s)
 X

x_i

Admissible data and
representations. $\mathcal{T}$

Input Manager

$\mathcal{IM}$

Transform and validate

$\downarrow x_i, f_k$

Machine Learning Models
 $A = \{ADM_1, \dots, ADM_K\}$

$\downarrow o_{ik}, \text{Performance}_k$

Decision Integration Layer

$\mathcal{OM}$

$\text{Performance}_k \rightarrow w_k, o_{ik} = ADM_k(x_i)$

$$d_i^{\text{final}} = \sum_k w_k o_{ik}$$

$\downarrow d_i^{\text{final}}$

Contextual Modulator

$\mathcal{C}$

$$c_i^{\text{final}} = \Lambda_{\text{final}}(\mu_1(x_i), \dots, \mu_L(x_i))$$

$\downarrow$

$$p_i := P(x_i) = \Gamma(d_i^{\text{final}}, c_i^{\text{final}})$$

$\downarrow p_i$

Prioritization ranking

π_P

Robustness

$\mathcal{R}$

$$Q_{\text{mod}} = \{q = (\gamma, \theta_\gamma), \gamma \in \{\Gamma_{\text{linear}}, \dots\}\}$$

$\downarrow \mathbf{p}^{(q)}$

$$\Pi_Q = \{\pi^{(q)} : q \in Q_{\text{mod}}\}$$

$\downarrow$

Robustness assessment

Explainability

$\mathcal{E}$

$$E_i^{\text{pred}} = \{(f, \Delta_{if})\}_{f \in \mathcal{F}}$$

$\downarrow$

$$E_i^{\text{context}} = \{(r, \mu_r^*(x_i))\}_{r \in R_a} \cup \{(d, \mu_d(x_i))\}_{d \in R_d}$$

$\downarrow$

$\downarrow$

$$E_i = \Psi(p_i, d_i^{\text{final}}, c_i^{\text{final}}, E_i^{\text{pred}}, E_i^{\text{context}}, \Gamma)$$

$\downarrow$

NLP Explanation

μ_r, μ_d, f_C

p_i

p_i